



Software engineer with expertise in scientific computing, simulation software, CI/CD infrastructure, numerical methods, and large-scale collaborative software development. Experienced building high-performance analysis tools, modernizing legacy systems, and developing reproducible workflows in C++ and Python.

Languages + Frameworks**Tools**

- | | |
|--|--------------------------------|
| • C++17/20 [STL, boost, HDF5, CxxTest] | • Git, Github Actions CI/CD |
| • Python [Pydantic, NumPy, FastAPI] | • GNU/Linux, GCC, Make, VSCode |
| • Bash, Fortran, R | • PyTest, Git LFS Mounts |

Scientific Software Engineer at Oak Ridge National Laboratory

Mar 2023 – present

- Technical release manager, gatekeeper, and contributor for Mantid, a 1.5Mloc, multi-institution, international software project.
- Develop and maintain large-scale C++17/20 and Python scientific software used for neutron-scattering data analysis at the Spallation Neutron Source.
- Refactored legacy HDF5 file-loading with strangler pattern, improving load performance 60× and removing external dependencies, while preserving backward compatibility.
- Contributor and senior member on SNAPRed, a data reduction code for highly-reconfigurable instruments, providing start-to-finish data organization and scientific workflows to SNS clients.
- Implement CI/CD workflows and automated testing infrastructure using GitHub Actions, CMake, and PyTest or CxxTest.
- Diagnose and resolve performance, numerical-correctness, and integration issues.
- Collaborate with instrument scientists, software engineers, and domain experts in distributed development teams.

Quant Researcher at [Anchorage Digital](#)

Oct 2022 - Mar 2023

- Analyze cryptocurrency market liquidity with Pandas, GSheets, and BigQuery.

Data Scientist at [Community](#)

Sep 2021 - Jun 2022

- Analyze big data for product insights in Jupyter with Pandas, NumPy, spaCy, and SKLearn.

Research Assistant at University of North Carolina - Chapel Hill

Aug 2016 – May 2022

- Designed, developed, tested, and maintained high-performance scientific software in C++ within Linux/GNU development environments.
- Implemented low-level numerical functionality using C++14, Bash, GCC, and Make-based build workflows.
- Integrated modern C++ software with legacy and contemporary Fortran scientific codebases.
- Developed software across the full lifecycle including architecture, implementation, validation, debugging, documentation, and performance analysis.
- Mentored high school and undergraduate research students in replicating numerical scientific workflows.

Ph.D., Physics University of North Carolina - Chapel Hill*Newtonian and Relativistic White Dwarf Asteroseismology*, 2022

Publications in MNRAS, Astrophysical Journal, Physical Review C and D, and SoftwareX.

